Project Proposal

IT 207 Human Intelligence

Problem and rationale

Intersections with changing approach demand can allocate green time inefficiently. This project continues an existing software prototype and evaluates whether simple adaptive controllers can respond to lane demand while preserving minimum service and fairness.

Aim and objectives

The aim is to build and evaluate a software-only traffic management system inspired by human attention, memory, learning, and decision-making. Objectives: retain the existing video pipeline; analyze lane-wise counts and queue proxies; expose recommendations and comparisons through an existing dashboard; and compare fixed-time, rule-based, and AI-assisted control under reproducible scenarios.

Scope and method

Recorded video is the intended perception source. YOLO vehicle detection and ByteTrack identities feed ROI counts, traffic analysis, and optional forecasts. A weighted controller combines current demand, waiting urgency, fairness, and predictions. The present experiment uses a seeded software queue simulator because SUMO and its dependencies are not available. No hardware or physical signal connection is in scope.

Evaluation and expected evidence

The prototype compares mean queue delay, mean queue length, throughput, and congestion rate. Detection precision, recall, mAP, tracking IDF1/MOTA, and forecasting MAE/RMSE require labeled data and trained models and will be reported only after those inputs are available. The current single-run simulation is preliminary, not evidence of real-world effectiveness.

Risks and ethics

Video may contain identifying information. A future study should minimize retention, limit access, state lawful purpose, and evaluate errors across lighting, camera, vehicle class, and traffic conditions. Recommendations remain advisory in software. The controller is not validated for public-road deployment.

Schedule

1. Audit and repair the supplied architecture. 2. Verify modules and the local simulator. 3. Collect or select licensed data and train/evaluate models. 4. Conduct repeated baseline experiments. 5. Finalize report, presentation, and viva material. Steps requiring absent datasets or system dependencies remain open.